Multivariable calculus · Differentiation in two variables

From Partial Derivatives to Implicit Curves

A complete worked lesson with diagrams, derivations and retrieval practice

Original instructional source included as sources/lesson.md. This is a demonstration, not an examination-frequency claim. Generated 17 October 2026.

· How to use this pack

Read each idea, follow the worked example, and reproduce the method without looking. Use the three graphs to connect symbols with slopes. Attempt the question bank before opening its worked answers. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

S1	Lesson: partial derivatives
S2	Lesson: total derivative
S3	Lesson: implicit curves

Contents

1. 1 · Partial derivatives and directional slopes
2. 2 · Total change and the chain rule
3. 3 · Implicit curves and valid tangent formulas
4. Appendix A · Symbol glossary
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1 Partial derivatives and directional slopes

Holding one input fixed gives a slope through a surface. This section distinguishes that coordinate slope from the change in an arbitrary unit direction, and explains when mixed derivatives agree. S1

Key terms and definitions

Partial derivative. The derivative with respect to one coordinate while all other independent coordinates remain fixed. S1

Unit direction. A vector of length one. Normalise a supplied direction before using $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$. S1

Formulas

E1.1 · Coordinate derivatives

$$f_x = 2xy, \qquad f_y = x^2 + \cos y$$

GIVES Coordinate slopes for $f = x^2y + \sin y$.

APPLIES Hold the other independent coordinate fixed.

UNITS Output/input; dimensionless in this example.

S1

E1.2 · Directional slope

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

GIVES Slope in a selected direction.

APPLIES f differentiable and $\|\mathbf{u}\| = 1$.

UNITS Output per unit distance.

S1

E1.3 · Second partials

$$f_{xx} = 2y, \quad f_{xy} = f_{yx} = 2x$$
$$f_{yy} = -\sin y$$

GIVES Variation of coordinate slopes; these are second partial derivatives.

APPLIES For the stated smooth function.

UNITS Output/input squared.

S1

Graphs and diagrams

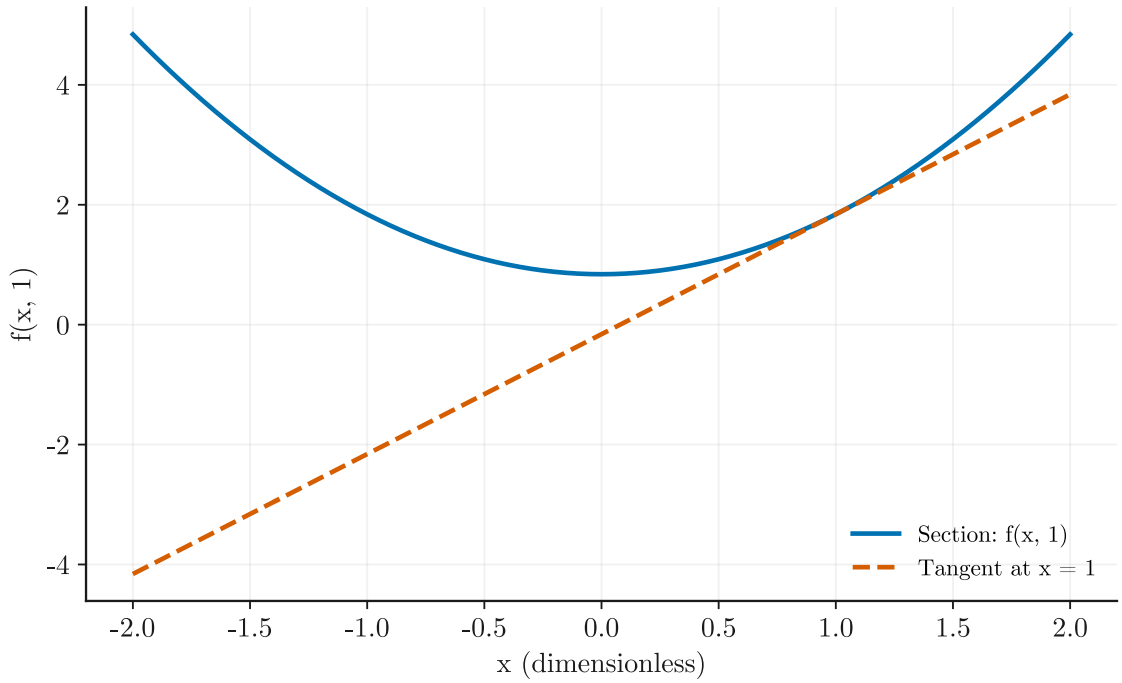


Figure G1.1. A coordinate section and its tangent.

S1

How to read it. Hold y=1 throughout the plot. The derivative with respect to x at x=1 is the slope of the dashed line.

Flowcharts

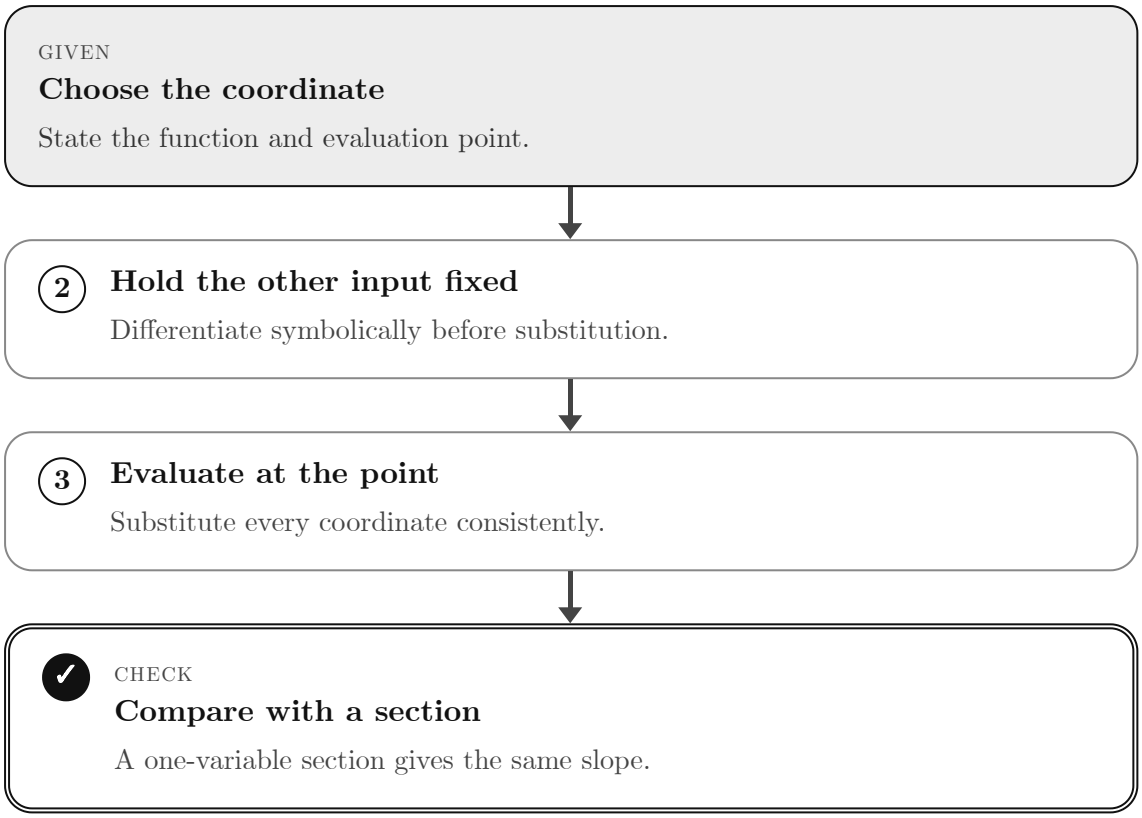


Figure F1.1. Evaluate a coordinate partial

S1

Important points and why

1. Differentiate before substituting the evaluation point. Substitution too early can erase the variable that must change. S1

Why can the same surface have two coordinate slopes?
An x-section and a y-section follow different curves through the same point, so their slopes need not agree.
S1

Exam trap. The gradient is a vector. A directional slope is the dot product of that vector with a unit direction. S1

Worked examples

W1.1 · Read two coordinate slopes

given	$f = x^2 y + \sin y$ at $(1, 0)$.
method	Coordinate differentiation
step 1	Hold y fixed to get $f_x = 2xy$. Hold x fixed to get $f_y = x^2 + \cos y$.
step 2	Substitute the point only after differentiating: $f_x(1,0) = 0$ and $f_y(1,0) = 1 + 1$.
answer	$\boxed{f_x(1,0) = 0,\ f_y(1,0) = 2.}$
units	Dimensionless coordinates.
check	The x-section at y=0 is identically zero. The y-section has slope 2 at the point.

S1

W1.2 · Choose an oblique direction

given	Use $\nabla f(1,0) = (0, 2)$ and $\mathbf{u} = (3/5, 4/5)$.
method	Directional derivative
step 1	Check the direction length: $(3/5)^2 + (4/5)^2 = 1$.
step 2	Take the dot product: $(0, 2) \cdot (3/5, 4/5) = 0 + 8/5$.
answer	$\boxed{D_{\mathbf{u}} f = 8/5.}$
units	Dimensionless coordinates.
check	The direction is a unit vector; no further scaling is needed.

S1

Practice with explained answers

Q1.1 · A claim about gradients

True or false: the gradient itself is the scalar slope in every direction.

S1

Answer · Q1.1

1. A gradient contains one component per independent coordinate.
2. Choose a unit direction and take a dot product to obtain its scalar slope.

Final answer. False. Use the directional derivative.

Check. Compare the x and y slopes in W1.1.

SHORT-NOTE RECAP

A partial derivative changes one coordinate while holding the others fixed. It is the slope of the corresponding surface section. The gradient collects these coordinate slopes, and its dot product with a unit vector gives the directional derivative. Mixed partials agree under suitable continuity assumptions.

S1

2 Total change and the chain rule

When both inputs change, their contributions combine. The total differential estimates a small finite change, while the chain rule gives the exact derivative of the composed function along a differentiable path. S2

Key terms and definitions

Total differential. The linear part of the change in a differentiable function, $df = f_x\,dx + f_y\,dy$. S2

Composed function. The one-variable function obtained after replacing every changing input by its path expression. S2

Formulas

E2.1 · Differential

$$df = f_x\,dx + f_y\,dy$$

GIVES	First-order change.
APPLIES	The function is differentiable at the base point. A finite increment is approximated by this differential.
UNITS	Same as the output.

S2

E2.2 · Chain rule

$$\frac{dz}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt}$$

GIVES	Change along a parameterised path.
APPLIES	Differentiate every changing input. Evaluate partials on the path.
UNITS	Output per parameter unit.

S2

Graphs and diagrams

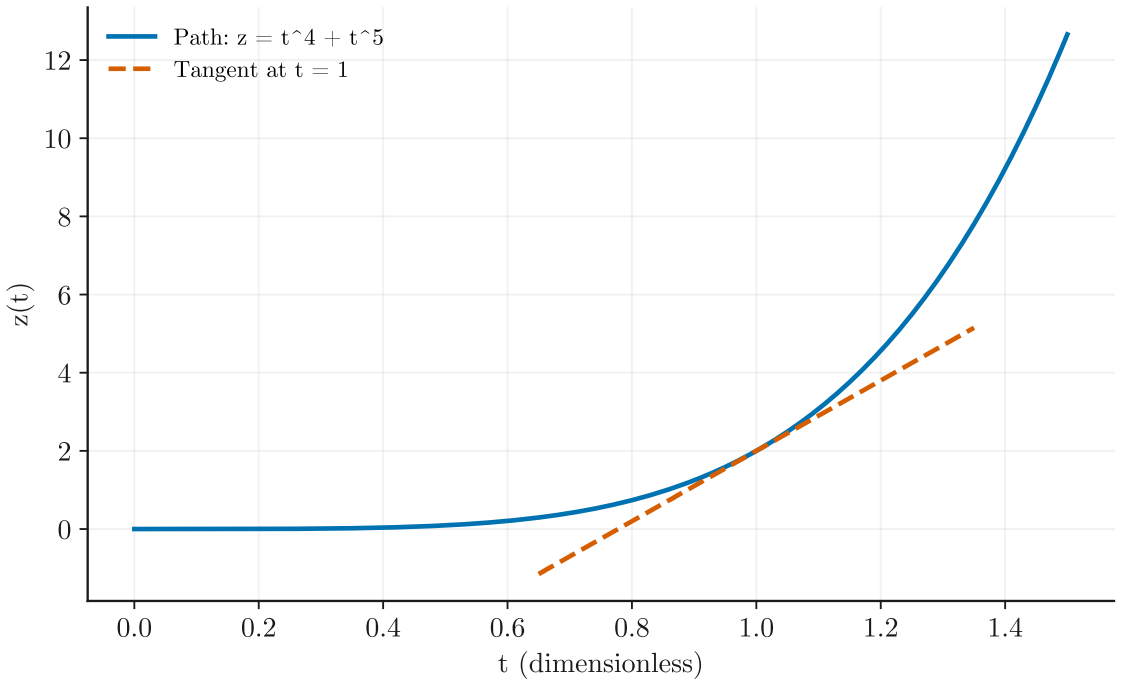


Figure G2.1. A changing path and its local rate.

S2

How to read it. The slope 9 belongs to change along the specified path, with both original inputs varying.

Flowcharts

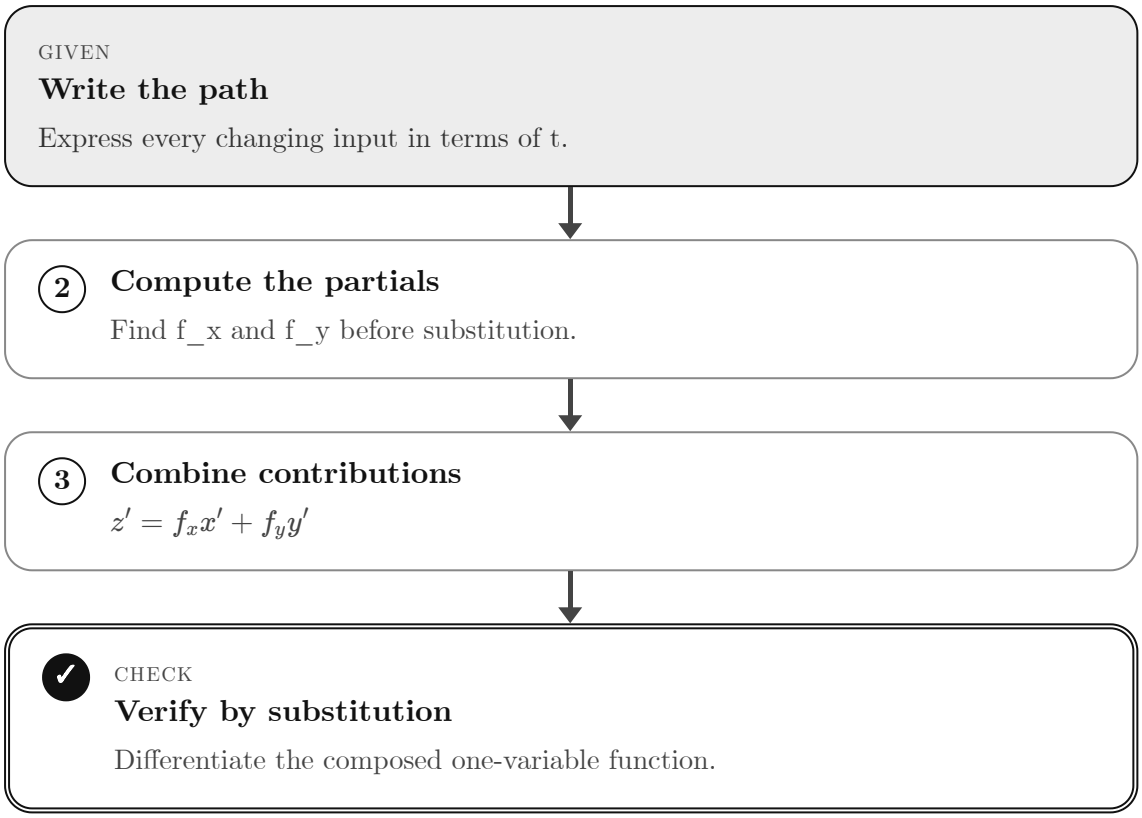


Figure F2.1. Account for both changing inputs

S2

Three related operations

Operation	What changes	What it returns
Partial derivative	One input; others fixed	Coordinate slope
Total differential	Small coordinate increments	Linear output change
Derivative along a path	All inputs specified by a parameter	Exact rate of the composition

S2

Important points and why

1. A path can change several inputs at once; include one chain-rule term for each changing input. S2

Why does a differential differ from a finite change?
The differential keeps the linear part. Nonlinear terms remain in the exact finite increment.
S2

Exam trap. Do not write a finite-change approximation as an exact equality. State the base point and the increments. S2

Worked examples

W2.1 · Differentiate along a moving path

GIVEN	$z = x^2y$, $x = t^2$, $y = 1 + t$; evaluate at $t = 1$.
METHOD	Two-path chain rule
STEP 1	Compute $z_x = 2xy$, $z_y = x^2$, $x' = 2t$ and $y' = 1$.
STEP 2	Use both paths: $z' = 2xy(2t) + x^2$.
STEP 3	At $t = 1$, $(x, y) = (1, 2)$, so $z' = 2(1)(2)(2) + 1$.
ANSWER	$\boxed{z'(1) = 9.}$
UNITS	Dimensionless coordinates.
CHECK	Direct substitution gives $z = t^4 + t^5$ and $z'(1) = 4 + 5 = 9$.

S2

W2.2 · Estimate a finite change

GIVEN	$f = x^2y + \sin y$ at $(1, 0)$ with $dx = 0.02$, $dy = -0.01$.
METHOD	First-order approximation
STEP 1	The gradient at the base point is $(0, 2)$.
STEP 2	The differential is $df = 0(0.02) + 2(-0.01)$.
STEP 3	Direct evaluation gives $\Delta f \approx -0.0204038333$; compare it with the linear estimate.
ANSWER	$\boxed{df = -0.02.}$
UNITS	Dimensionless coordinates.
CHECK	The absolute discrepancy is about 0.0004038333, which is a higher-order remainder.

S2

Practice with explained answers

Q2.1 · A missing chain-rule path

A student differentiates $z = x^2y$ along $x = t^2$, $y = 1 + t$ and writes $z' = 2xy(2t)$. Find and repair the omission.

S2

Answer · Q2.1

1. The student included the x path but omitted the y path.
2. Add $z_y y' = x^2$. Then substitute the path.

Final answer. $z' = 4t^3 + 5t^4$; at $t = 1$, the rate is 9.

Check. Direct substitution gives z=t^4+t^5.

short-note recap

For a differentiable function, the total differential combines the coordinate contributions to first-order change. Along a differentiable path, the chain rule gives the exact derivative of the composition. Check the result by direct substitution when that route is practical.

S2

3 Implicit curves and valid tangent formulas

An equation can define a curve without giving y explicitly. Differentiate the relation, check that the point lies on the curve, and confirm the denominator is nonzero before evaluating a slope. S3

Key terms and definitions

Implicit relation. An equation $F(x, y) = 0$ that describes admissible pairs (x, y). S3

Regular branch. A local representation of the curve as a differentiable function $y(x)$, justified here by $F_y \neq 0$. S3

Formulas

E3.1 · Implicit first derivative

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

gives	Slope of a regular implicit branch.
applies	$F = 0$ at the point and $F_y \neq 0$.
units	Units of y divided by units of x.

S3

E3.2 · Circle second derivative

$$y'' = -\frac{25}{y^3}$$

gives	Second derivative of a branch of $x^2 + y^2 = 25$.
applies	$y \neq 0$; this formula is specific to the radius-five circle.
units	Dimensionless coordinates.

S3

Derivations

H3.1 · Derive the second derivative on the circle

given	$x^2 + y^2 = 25$, with $y \neq 0$.
step 1	Differentiate once: $2x + 2yy' = 0$, so $y' = -x/y$.
step 2	Differentiate again using the product rule: $2 + 2(y')^2 + 2yy'' = 0$.
step 3	Solve for $y'' = -(1 + (y')^2)/y$.
step 4	Substitute $y' = -x/y$ and use $x^2 + y^2 = 25$.
answer	$\boxed{y'' = -25/y^3}$; at $(3, 4)$ it is $-25/64$.
check	On the upper semicircle y is positive, so the second derivative is negative, as its concave-down shape requires.

S3

Graphs and diagrams

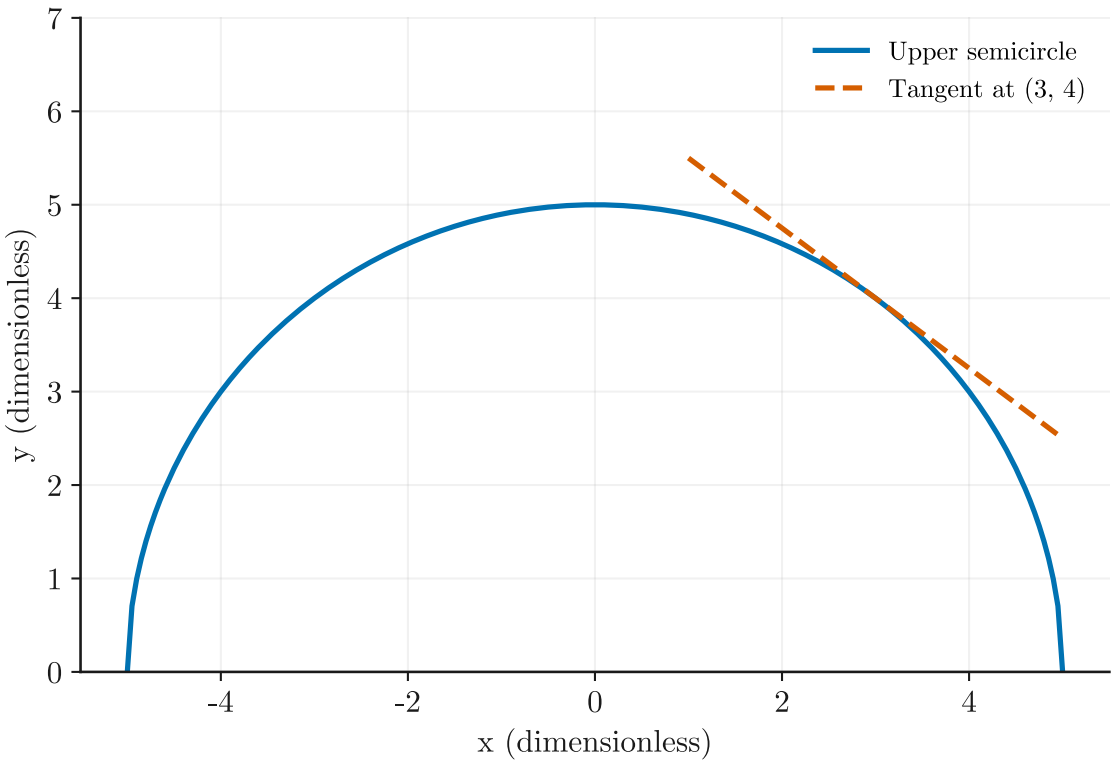


Figure G3.1. A circle branch and its tangent.

S3

How to read it. The tangent is perpendicular to the radius at the contact point. Its downward slope agrees with the implicit derivative.

Flowcharts

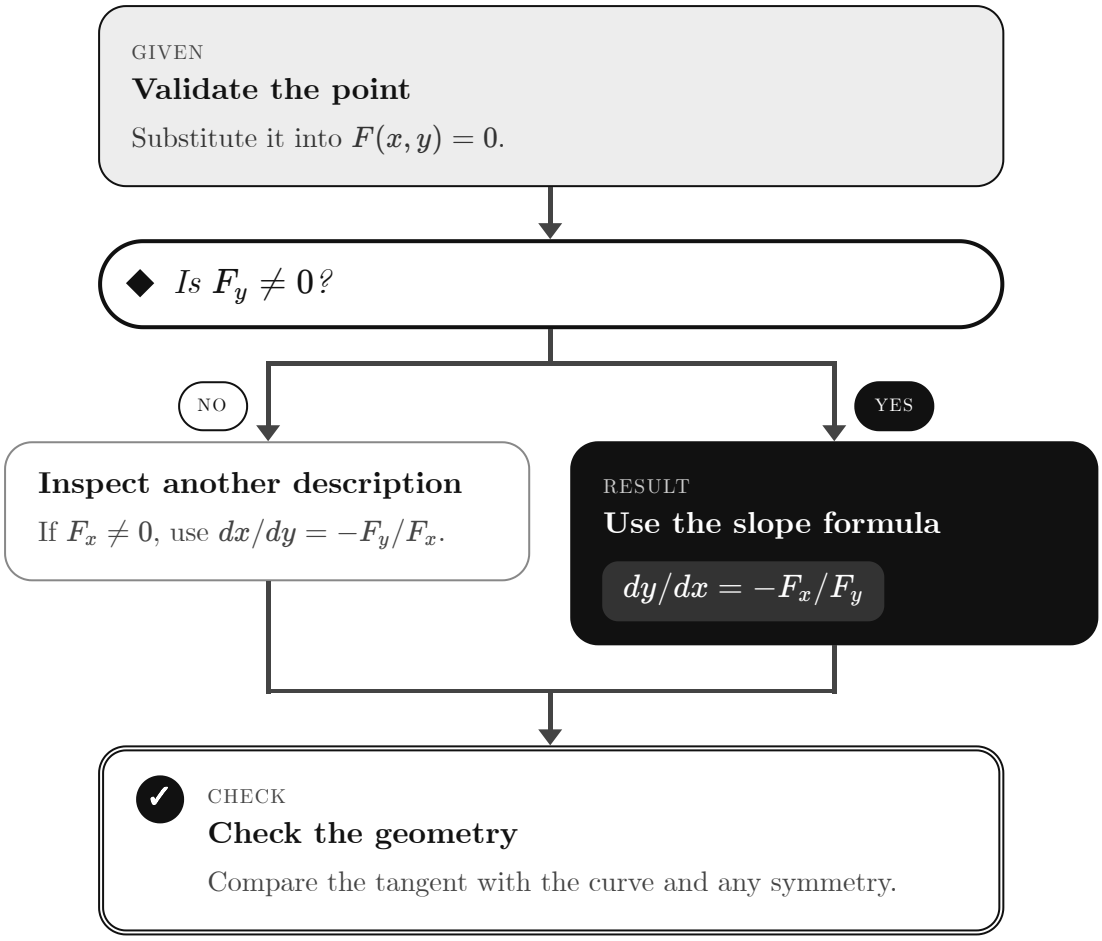


Figure F3.1. Check the implicit-derivative denominator

S3

Important points and why

1. Verify the point belongs to the relation and the chosen denominator is nonzero before evaluating an implicit derivative. S3

Why does a zero denominator require a new analysis?
It prevents division in the chosen formula. The curve may instead have a vertical tangent or fail to be regular; examine the defining equation.
S3

Exam trap. Do not evaluate a tangent at a point that is not on the curve. Do not turn division by zero into a slope of zero. S3

Worked examples

W3.1 · Find the tangent to a circle

GIVEN	$x^2 + y^2 = 25$ at $(3, 4)$.
METHOD	Implicit differentiation
STEP 1	Validate the point: $3^2 + 4^2 = 25$.
STEP 2	Differentiate: $2x + 2y\,y' = 0$. Since $2y = 8 \neq 0$, divide safely.
STEP 3	Evaluate $y' = -x/y = -3/4$ and use the point-slope equation.
ANSWER	$\boxed{y - 4 = -\frac{3}{4}(x - 3).}$
UNITS	Dimensionless coordinates.
CHECK	The radius vector $(3, 4)$ is perpendicular to tangent direction $(4, -3)$ because their dot product is zero.

S3

W3.2 · Recognise a vertical tangent

GIVEN	$x^2 + y^2 = 25$ at $(5, 0)$.
METHOD	Domain and regularity check
STEP 1	Check $5^2 + 0^2 = 25$, so the point lies on the curve.
STEP 2	The denominator $F_y = 2y$ is zero; the formula for dy/dx cannot be evaluated.
STEP 3	Instead use $dx/dy = -y/x = 0$, since $F_x = 10 \neq 0$.
ANSWER	$\boxed{\text{The tangent is the vertical line } x = 5.}$
UNITS	Dimensionless coordinates.
CHECK	Changing the dependent variable gives a regular local description; a zero denominator is not a zero slope.

S3

Practice with explained answers

Q3.1 · Circle second derivative

Find y'' on $x^2 + y^2 = 25$ at $(3, 4)$.

S3

Answer · Q3.1

1. Use the circle formula $y'' = -25/y^3$.
2. Substitute $y = 4$ after checking the point and denominator.

Final answer. $-25/64$.

Check. The negative second derivative agrees with the concave-down upper semicircle.

short-note recap

Implicit differentiation applies the chain rule to a relation. For a regular branch with nonzero F_y, the slope is minus F_x divided by F_y. The equation itself validates the evaluation point. A zero denominator may indicate that x should be treated as a function of y instead.

S3

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
f_x	Partial derivative in x	Change in f per change in x while y stays fixed. S1	output/x	E1.1
u	Unit direction	Chosen vector of length one. S1	dimensionless	E1.2
df	Differential	Linear part of output change. S2	output	E2.1
t	Path parameter	Independent variable along a path. S2	parameter unit	E2.2

Symbol	Name	Meaning	Unit	Appears in
F	Defining function	The relation is the level set F=0. S3	context-dependent	E3.1
x, y	Independent coordinates	The two real input coordinates of f. S1	Dimensionless	E1.1
f	Scalar function	In this lesson, $f(x, y) = x^2 y + \sin y$. S1	Dimensionless	E1.1
∇f	Gradient	The vector (f_x, f_y); its dot product with a unit vector gives the directional derivative. S1	Output per input	E1.2
$D_{\mathbf{u}} f$	Directional derivative	The rate of change per unit displacement in the unit direction $\mathbf{u}$. S1	Output per input	E1.2
y', y''	Implicit derivatives	First and second derivatives of a local branch $y(x)$, where that branch exists. S3	Output per input; output per input squared	E3.1, E3.2

B Sources, coverage and gaps

Source	Path	Use
Original derivative lesson	sources/lesson.md	Theory, examples and all illustrative plots

No past papers supplied. All practice questions are authored examples. No ESA ranking is asserted.

Known gaps, conflicts and corrections

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
S1	SRC1/partial	covered	Read the full partial section; mapped its definitions, equations, conditions, examples, pitfalls and geometry.
S2	SRC1/total	covered	Read the full total section; mapped its definitions, equations, conditions, examples, pitfalls and geometry.
S3	SRC1/implicit	covered	Read the full implicit section; mapped its definitions, equations, conditions, examples, pitfalls and geometry.

C Revision and retrieval practice

Keywords

Gradient. Vector of coordinate partial derivatives. S1

Directional derivative. Rate along a specified unit direction. S1

Mixed partial. Derivative taken with respect to different coordinates. S1

Differential. Linear change at a specified base point. S2

Chain rule. Derivative of a composition through all changing inputs. S2

Remainder. Difference between an exact increment and a chosen approximation. S2

Implicit function. Local function represented by a relation. S3

Regular point. Here, a point where the gradient of the relation is nonzero. S3

Vertical tangent. Tangent parallel to the y-axis. S3

Rapid recall

Why can the same surface have two coordinate slopes? An x-section and a y-section follow different curves through the same point, so their slopes need not agree. S1

Why does a differential differ from a finite change? The differential keeps the linear part. Nonlinear terms remain in the exact finite increment. S2

Why does a zero denominator require a new analysis? It prevents division in the chosen formula. The curve may instead have a vertical tangent or fail to be regular; examine the defining equation. S3

Essential facts

Differentiate before substituting the evaluation point. Substitution too early can erase the variable that must change. S1

A path can change several inputs at once; include one chain-rule term for each changing input. S2

Verify the point belongs to the relation and the chosen denominator is nonzero before evaluating an implicit derivative. S3